

PADTHAWAY SOUTH, Australia

WMO: 958230

Lat: 36.6539S

Lon: 140.5211E

Elev: 121

StdP: 14.63

Time Zone: 9.50 (AUA)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.9	35.2	28.1	22.2	67.3	31.0	25.3	55.3	24.3	53.3	22.5	55.2	2.1	350	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.7	98.6	65.8	93.9	64.7	89.8	63.7	69.5	85.6	67.6	84.6	66.1	83.1	11.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	93.9	72.3	62.7	85.7	71.3	60.6	79.2	69.5	33.6	85.0	32.0	84.4	30.8	83.1	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.8	17.0	29.0	107.3	1.9	3.0	27.6	109.4	26.4	111.2	25.4	112.8	24.0	115.0		
(5)			28.5	72.6	2.0	1.5	27.1	73.7	25.9	74.6	24.8	75.4	23.3	76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	68.7	68.1	64.5	58.9	53.9	49.8	49.4	50.5	53.4	57.1	61.7	65.3		
	DBStd	9.42	7.93	7.61	7.58	6.28	5.26	4.32	3.98	4.55	5.41	7.20	8.03	7.72		
	HDD50	214	0	0	0	2	18	56	62	49	18	8	1	0		
	HDD65	3011	42	46	105	206	346	457	484	450	353	268	163	91		
	CDD50	3275	580	507	449	268	140	50	43	64	118	229	351	476		
	CDD65	595	157	133	89	22	3	0	0	0	3	24	63	101		
	CDH74	8797	2386	1804	1225	291	10	0	0	2	34	382	983	1680		
	CDH80	4239	1269	892	558	87	1	0	0	0	2	138	443	849		
(14) Wind	WSAvg	8.2	8.5	8.1	7.6	6.8	7.4	7.3	9.0	8.9	8.6	8.7	8.6	8.5		
(15) Precipitation	PrecAvg	19.5	1.0	0.8	0.8	1.1	1.8	2.6	2.8	2.4	2.2	1.7	1.1	1.2		
	PrecMax	26.4	3.1	2.2	2.2	2.8	4.3	4.3	4.7	4.8	4.6	4.0	2.7	4.4		
	PrecMin	12.0	0.1	0.1	0.1	0.1	0.2	0.5	1.1	0.3	0.5	0.3	0.3	0.2		
	PrecStd	3.5	0.7	0.6	0.5	0.7	0.9	1.0	0.9	1.2	1.1	1.0	0.7	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	106.5	100.5	98.6	88.8	74.7	65.4	63.7	71.2	78.4	91.0	94.6	102.2		
		MCWB	67.8	66.7	65.0	62.9	57.0	54.7	54.0	55.7	58.8	60.9	65.1	65.3		
	2%	DB	99.4	95.3	91.9	81.3	69.8	61.5	60.2	65.8	72.4	83.5	90.9	96.1		
		MCWB	66.8	66.1	64.7	60.1	56.7	53.8	52.7	54.2	56.0	59.0	62.8	63.7		
	5%	DB	92.8	90.8	86.6	76.6	66.0	59.4	58.2	61.4	67.8	77.3	85.4	89.6		
		MCWB	65.2	65.6	63.0	58.5	56.0	53.1	51.9	52.6	54.7	58.0	62.3	62.4		
	10%	DB	86.8	85.3	80.7	71.8	62.9	57.5	56.4	58.5	63.8	72.4	79.1	83.4		
		MCWB	63.7	64.0	61.4	57.7	55.2	52.3	51.1	51.7	53.8	56.8	60.5	61.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.6	72.6	68.7	64.5	61.1	57.8	56.1	57.5	59.9	64.4	69.2	70.4		
		MCDB	88.3	92.2	82.9	77.5	66.4	61.6	59.8	66.6	72.5	79.5	84.8	84.9		
	2%	WB	69.6	69.1	66.5	62.6	59.2	55.4	54.3	55.4	57.9	61.4	66.5	66.6		
		MCDB	86.5	85.8	83.1	73.5	65.0	59.3	58.1	62.4	67.9	75.9	82.0	83.8		
	5%	WB	67.5	67.2	64.9	60.8	57.6	54.0	53.1	53.8	56.3	59.5	64.3	64.8		
		MCDB	87.1	83.1	80.8	71.2	63.2	57.9	56.8	59.9	64.8	73.6	80.1	82.9		
	10%	WB	65.4	65.2	63.4	59.2	56.2	52.8	51.9	52.3	54.6	57.7	61.9	63.0		
		MCDB	84.0	80.6	78.3	68.8	61.7	56.6	55.3	57.5	62.3	69.9	75.6	80.6		
(35) Mean Daily Temperature Range	MDBR		28.7	27.5	26.1	23.5	17.7	15.8	14.0	15.7	18.6	23.5	25.8	27.8		
	5% DB	MCDBR	38.4	34.6	33.6	30.5	20.9	17.0	14.8	19.2	24.3	32.4	34.3	39.4		
		MCWBR	11.5	11.2	11.2	12.8	11.4	11.2	9.6	10.9	11.4	13.4	12.2	12.3		
	5% WB	MCDBR	30.0	27.8	27.7	24.0	16.7	14.7	12.6	16.9	20.6	27.2	29.2	29.4		
		MCWBR	11.2	10.6	10.9	11.9	10.4	10.4	8.9	10.5	11.1	12.7	11.9	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.342	0.335	0.319	0.313	0.302	0.298	0.299	0.313	0.332	0.348	0.338	0.334		
	taud		2.476	2.512	2.555	2.555	2.573	2.570	2.562	2.503	2.440	2.396	2.463	2.494		
	Ebn at Noon		315	310	303	285	270	262	268	281	293	302	314	318		
	Edh at Noon		37	34	30	26	23	21	23	27	33	38	37	36		
(44) All-Sky Solar Radiation	RadAvg		2420	2124	1675	1181	795	650	698	975	1344	1783	2127	2373		
	RadStd		128	108	68	71	40	48	30	50	107	129	127	122		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (18 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-182	+247	+74	

Nomenclature: See separate page